Today

- General angular momentum
- Raising and lowering operators
- Spin
- Total angular momentum
- Addition of angular momentum

4.5 General angular momentum

Now we want to investigate angular momentum based on the commutation relations and without consideration of representation either in position or momentum space. We call such angular momentum general angular momentum and denote it by $\hat{\vec{J}}$. In terms of its cartesian components

$$\hat{\vec{J}} = (\hat{J}_x, \hat{J}_y, \hat{J}_z) = (\hat{J}_1, \hat{J}_2, \hat{J}_3), \tag{4.5.1}$$

where we interchangeably use the indices $1, 2, 3$ for x, y, z. For the total angular momentum we write

$$\hat{\vec{J}}^2 = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2. \tag{4.5.2}$$

We take the commutation relations are

$$[\hat{J}_i, \hat{J}_j] = i\hbar\epsilon_{ijk}\hat{J}_k \tag{4.5.3}$$

as the definition of angular momentum. Moreover,

$$[\hat{\vec{J}}^2, \hat{J}_i] = 0. \tag{4.5.4}$$

We take the commuting set to be $\hat{\vec{J}}^2$ and $\hat{J}_z$. Now suppose we trade $\hat{J}_x$ and $\hat{J}_y$ for

$$\hat{J}_\pm = \hat{J}_x \pm i\hat{J}_y. \tag{4.5.5}$$

The commutation relations become

$$[\hat{J}_+, \hat{J}_-] = [\hat{J}_x + i\hat{J}_y, \hat{J}_x - i\hat{J}_y]$$

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$$= -i[\hat{J}_x, \hat{J}_y] + i[\hat{J}_y, \hat{J}_x] = -i(i\hbar\hat{J}_z) + i(-i\hbar\hat{J}_z) = 2\hbar\hat{J}_z. \tag{4.5.6}$$

$$\begin{aligned}[\hat{J}_z, \hat{J}_\pm] &= [\hat{J}_z, \hat{J}_x \pm i\hat{J}_y] \\ &= [\hat{J}_z, \hat{J}_x] \pm i[\hat{J}_z, \hat{J}_y] = i\hbar\hat{J}_y \pm i(i\hbar\hat{J}_x) = \pm\hbar(\hat{J}_x \pm i\hat{J}_y) = \pm\hbar\hat{J}_\pm.\end{aligned} \tag{4.5.7}$$

Moreover, because, $\hat{\vec{J}}^2$ commutes with every components of $\hat{\vec{J}}$,

$$[\hat{\vec{J}}^2, \hat{J}_\pm] = 0. \tag{4.5.8}$$

Because $\hat{\vec{J}}^2$ and $\hat{J}_z$ commute, we can simultaneously diagonalize them. Let the simultaneous eigenstate be $|ab\rangle$ such that

$$\begin{aligned}\hat{\vec{J}}^2|ab\rangle &= a|ab\rangle \\ \hat{J}_z|ab\rangle &= b|ab\rangle\end{aligned} \tag{4.5.9}$$

Question 4.5.1: Which of the following statements are true? (A). $\hat{J}_+$ raises the eigenvalue a. (B). $\hat{J}_+$ raises the eigenvalue b. (C). $\hat{J}_-$ lowers the eigenvalue a. (D). $\hat{J}_-$ lowers the eigenvalue b.

First,

$$\hat{\vec{J}}^2\left(\hat{J}_\pm|ab\rangle\right) = \left([\hat{\vec{J}}^2, \hat{J}_\pm] + \hat{J}_\pm\hat{\vec{J}}^2\right)|ab\rangle = \left(0 + a\hat{J}_\pm\right)|ab\rangle = a\left(\hat{J}_\pm|ab\rangle\right). \tag{4.5.10}$$

Therefore, $\hat{J}_\pm|ab\rangle$ is an eigenstate of $\hat{\vec{J}}^2$ with the same eigenvalue a as $|ab\rangle$ and does not raise or lower a, as expected, since $[\hat{\vec{J}}^2, \hat{J}_\pm] = 0$. On the other hand,

$$\hat{J}_z\left(\hat{J}_\pm|ab\rangle\right) = \left([\hat{J}_z, \hat{J}_\pm] + \hat{J}_\pm\hat{J}_z\right)|ab\rangle = \left(\pm\hbar\hat{J}_z + b\hat{J}_\pm\right)|ab\rangle = (b \pm \hbar)\left(\hat{J}_\pm|ab\rangle\right). \tag{4.5.11}$$

Thus, $\hat{J}_+$ raises and $\hat{J}_-$ lowers the eigenvalue a of $\hat{J}_z$. Therefore, assuming $|ab\rangle$ is normalized,

$$\hat{J}_\pm|ab\rangle = c_\pm|ab \pm \hbar\rangle, \tag{4.5.12}$$

where $c_\pm$ are normalization constants.

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Note that

$$\begin{aligned}\hat{J}_{\pm}\hat{J}_{\mp} &= (\hat{J}_x \pm i\hat{J}_y)(\hat{J}_x \mp i\hat{J}_y) \\ &= \hat{J}_x^2 + \hat{J}_y^2 \mp i\hat{J}_x\hat{J}_y + \pm i\hat{J}_y\hat{J}_x = \hat{J}_x^2 + \hat{J}_y^2 \mp i[\hat{J}_x, \hat{J}_y] \\ &= \hat{J}_x^2 + \hat{J}_y^2 \pm \hbar\hat{J}_z = \hat{\vec{J}}^2 - \hat{J}_z^2 \pm \hbar\hat{J}_z, \end{aligned} \tag{4.5.13}$$

or

$$\hat{J}_{\mp}\hat{J}_{\pm} = \hat{J}_x^2 + \hat{J}_y^2 \mp \hbar\hat{J}_z = \hat{\vec{J}}^2 - \hat{J}_z^2 \mp \hbar\hat{J}_z, \tag{4.5.14}$$

and

$$\frac{1}{2}\left(\hat{J}_+\hat{J}_- + \hat{J}_-\hat{J}_+\right) = \hat{J}_x^2 + \hat{J}_y^2 = \hat{\vec{J}}^2 - \hat{J}_z^2. \tag{4.5.15}$$

In particular,

$$\hat{J}_-\hat{J}_+ = \hat{\vec{J}}^2 - \hat{J}_z^2 - \hbar\hat{J}_z, \qquad \hat{J}_+\hat{J}_- = \hat{\vec{J}}^2 - \hat{J}_z^2 + \hbar\hat{J}_z. \tag{4.5.16}$$

Now consider states with state $|ab_{\max}\rangle$ having the highest value of $b = b_{\max}$ such that $\hat{J}_+|ab_{\max}\rangle = 0$ and $|ab_{\min}\rangle$ having the lowest value of $b = b_{\min}$ such that $\hat{J}_-|ab_{\min}\rangle = 0$.

And then, taking the inner product of $\hat{J}_+|ab_{\max}\rangle$ with itself,

$$\begin{aligned} 0 &= \langle ab_{\max}|\hat{J}_-\hat{J}_+|ab_{\max}\rangle = \langle ab_{\max}|\hat{\vec{J}}^2 - \hat{J}_z^2 - \hbar\hat{J}_z|ab_{\max}\rangle \\ &= a - b_{\max}^2 - \hbar b_{\max}. \end{aligned} \tag{4.5.17}$$

In other words,

$$b_{\max} = \frac{-\hbar \pm \sqrt{\hbar^2 + 4a}}{2} \tag{4.5.18}$$

Furthermore, taking the inner product of $\hat{J}_-|ab_{\min}\rangle$ with itself,

$$\begin{aligned} 0 &= \langle ab_{\min}|\hat{J}_+\hat{J}_-|ab_{\min}\rangle = \langle ab_{\min}|\hat{\vec{J}}^2 - \hat{J}_z^2 + \hbar\hat{J}_z|ab_{\min}\rangle \\ &= a - b_{\min}^2 + \hbar b_{\min}. \end{aligned} \tag{4.5.19}$$

In other words,

$$b_{\min} = \frac{\hbar \pm \sqrt{\hbar^2 + 4a}}{2} \tag{4.5.20}$$

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Because we require $b_{\max} \geq b_{\min}$, we choose the plus sign in $b_{\max}$ and the minus sign in $b_{\min}$; i.e.,

$$b_{\max} = \frac{-\hbar + \sqrt{\hbar^2 + 4a}}{2}, \qquad b_{\min} = \frac{\hbar - \sqrt{\hbar^2 + 4a^2}}{2}. \tag{4.5.21}$$

Note that $b_{\min} = -b_{\max}$ Because the value of b increases in steps of $\hbar$, we must have

$$b_{\max} = b_{\min} + n\hbar, \tag{4.5.22}$$

where n is a non-negative integer, $n = 0, 1, 2, \ldots$ Moreover, because of isotropy of space the maximum value for the eigenvalue of $\hat{L}_z$ in one frame could be converted to a minimum value in another, or just using (4.5.21),

$$b_{\min} = -b_{\max}. \tag{4.5.23}$$

Thus

$$b_{\max} = -b_{\max} + n\hbar \tag{4.5.24}$$

or

$$b_{\max} = -b_{\min} = \frac{n}{2}\hbar, \qquad n = 0, 1, 2, \ldots \tag{4.5.25}$$

We define $j = \frac{n}{2}$ and rewrite this as

$$b_{\max} = -b_{\min} = j\hbar, \qquad j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \ldots \tag{4.5.26}$$

The value of a then follows from

$$a = b_{\max}^2 + \hbar b_{\max} = j^2\hbar^2 + \hbar^2 j = j(j+1)\hbar^2. \tag{4.5.27}$$

> **Question 4.5.2:** What important difference do you see between orbital and general angular momentum? Which of the following are correct regarding orbital and general angular momenta? (A). Total orbital angular momentum can take $j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$, (B). Total angular momentum can take $j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$, (C). Total orbital angular momentum can take only $j = 0, 1, 2, \ldots$, (A). Total angular momentum can take only $j = 0, 1, 2, \ldots$

Thus, for a given $b_{\max} = j\hbar$ or $a = j(j+1)\hbar^2$, the possible values of b range from $b_{\min} = -j\hbar$ to $b_{\max} = j\hbar$ in steps of $\hbar$. In other words, b takes values $m_j\hbar$, where

$$m_j = -j, -j+1, \ldots, j. \tag{4.5.28}$$

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To determine the normalization constants, let us first rewrite (4.5.12) as

$$\hat{J}_\pm|jm_j\rangle = c^\pm_{jm_j}|jm_j \pm 1\rangle, \tag{4.5.29}$$

The inner product is

$$\langle jm_j|\hat{J}_\mp\hat{J}_\pm|jm_j\rangle \quad = \quad |c^\pm_{jm_j}|^2\langle jm_j \pm 1|\hat{J}_\mp\hat{J}_\pm|jm_j \pm 1\rangle = |c^\pm_{jm_j}|^2 \tag{4.5.30}$$

On the other hand,

$$\langle jm_j|\hat{J}_\mp\hat{J}_\pm|jm_j\rangle \quad = \quad \langle jm_j|\hat{\vec{J}}^2 - \hat{J}_z^2 \mp \hbar\hat{J}_z|jm_j\rangle = j(j+1)\hbar^2 - m_j^2\hbar^2 \mp \hbar^2 m_j. \tag{4.5.31}$$

Therefore,

$$|c^\pm_{jm_j}|^2 = j(j+1)\hbar^2 - m_j^2\hbar^2 \mp \hbar^2 m_j = \hbar^2(j \mp m_j)(j \pm m_j + 1) \tag{4.5.32}$$

Choosing real normalization constants,

$$\hat{J}_\pm|jm_j\rangle \quad = \quad \hbar\sqrt{(j \mp m_j)(j \pm m_j + 1)}\,|jm_j \pm 1\rangle. \tag{4.5.33}$$

Note that the matrix elements of $\hat{J}_\pm$ are

$$\begin{aligned}\langle j'm_j'|\hat{J}_\pm|jm_j\rangle \quad &= \quad \hbar\sqrt{(j \mp m_j)(j \pm m_j + 1)}\langle j'm_j'|jm_j \pm 1\rangle \\ &= \quad \hbar\sqrt{(j \mp m_j)(j \pm m_j + 1)}\,\delta_{j'j}\delta_{m_j',m_j\pm1}.\end{aligned} \tag{4.5.34}$$

4.6 Spin

4.6.1 Classical orbital and spinning motion

The revolution of the earth around the sun is an example of a classical orbital motion. The rotation of the earth about an axis through it is an example of a spinning motion. In either case, the infinitesimal classical angular momentum corresponding to an infinitesimal linear momentum $d\vec{p} = dm\vec{v}$ of mass dm at position $\vec{r}$ from the axis of rotation moving with velocity $\vec{v}$ is given by

$$d\vec{L} = \vec{r} \times d\vec{p}. \tag{4.6.1}$$

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The total angular momentum is

$$\vec{L} = \int \vec{r} \times d\vec{p} = \int \vec{r} \times dm\vec{v}. \tag{4.6.2}$$

In particular, for a point particle of mass m at fixed radius r spinning with constant angular velocity ω about the $z-$axis,

$$\vec{L} = I\omega\hat{z} = m\omega r^2\hat{z}. \tag{4.6.3}$$

If the rotating particle has charge q, it also has a magnetic dipole moment $\vec{\mu}$. Together with the angular momentum $\vec{L}$,

$$\vec{L} = m\omega r^2\hat{z}, \qquad \vec{\mu} = \frac{q}{2}\omega r^2\hat{z}, \tag{4.6.4}$$

where we used $\mu = I\pi r^2$ with current $I = \frac{q}{2\pi/\omega} = \frac{q\omega}{2\pi}$. Thus,

$$\vec{\mu} = \frac{q}{2m}\vec{L}. \tag{4.6.5}$$

Exercise 4.6.1:

1. Consider a uniformly charged wire ring of mass M and total charge Q rotating about its axis. Find the ratio of its magnetic dipole moment to its angular momentum. This is called the gyromagnetic ratio. (5 points)
2. Generalize your result in part (1) for the gyromagnetic ratio of a uniformly charged spinning sphere of mass M carrying charge Q. (5 points)
3. Supposing that the electron is a uniformly charged sphere and using your result in part (2), calculate the numerical value of its gyromagnetic ratio. (5 points)
4. Experiment puts an upper bound of about 10^{-22} m to the radius of the electron. Supposing that the electron is a spherical shell of uniform mass density and assuming that the farthest point on the surface spins at the speed of light, find an upper bound to the classical angular momentum of the electron. Using this and the gyromagnetic ratio in part (3), find the corresponding numerical value of the magnetic dipole moment of the electron. (5 points)
5. Experiment shows that the electron has intrinsic angular momentum called spin with value of $\frac{\hbar}{2}$, where $\hbar = \frac{h}{2\pi}$ and h is Planck's constant, and that the gyromagnetic ratio is bigger by about a factor of 2 called the $g-$factor. The cur-

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rent experimental value of the electron spin $g-$factor is 2.00231930436256(35), and the experimental value of the magnetic dipole moment of the electron is $[-928.476377 \times 10^{-26} \pm (0.000023 \times 10^{-26})]JT^{-1}$. How does the classical angular momentum in part (4) compare to the experimental value of spin, and what does this tell us about the possibility of a classical interpretation to spin? (5 points)

Solution:

1. The magnitude of magnetic dipole moment is $\mu = Ia = \frac{Q}{T}\pi R^2 = \frac{Qv}{2\pi R}\pi R^2 = \frac{1}{2}QvR$ and the angular momentum is $L = MvR$. The gyromagnetic ratio is then $g = \frac{m}{L} = \frac{Q}{2M}$.
2. We can decompose the sphere into rings with infinitesimal thickness and, because g does not depend on the radius of the rings, the gyromagnetic ratio is $g = \frac{Q}{2M}$.
3. For the electron $g = \frac{-e}{2m_e} = -8.8 \times 10^{10} C.kg^{-1}$.
4. The angular momentum is $L = I\omega = \frac{2}{3}m_e r_e^2 \frac{c}{r_e} = \frac{2}{3}m_e r_e c \leq \frac{2}{3}(9.11 \times 10^{-31} kg)(10^{-22} m)(3 \times 10^{-8} m.s^{-1}) = 1.8 \times 10^{-44} J.s^{-1}$. The magnetic dipole moment is $\mu = gL \leq 1.6 \times 10^{-33} J.T^{-1}$.
5. The classical value of angular momentum is much smaller than the measured value, $\frac{L}{\hbar/2} \leq 3.5 \times 10^{-10}$. Thus the spin of the electron cannot be explained by a classical spinning motion. [Aside remark: Spin is one of the defining intrinsic properties of elementary particles, just like electric charge and mass.]

4.6.2 Spin

Spin is a quantum mechanics property with no classical analogue. In particular, as we see in Ex. 4.6.1, the magnitude of the intrinsic angular momentum of an electron is too large to be reproduced by a classical spinning motion. A measurement of the spin of an electron along a given direction, such as along z, could produce either one of only two possible values. Thus, the electron has $j = \frac{1}{2}$ and $m_j = \pm\frac{1}{2}$.

In other words, the special case of $j = \frac{1}{2}$ describes the spin (or spin angular momentum) of a particle of spin$-\frac{1}{2}$, and we often denote it by $s = \frac{1}{2}$. We also often denote the spin angular momentum operator by $\hat{S}$ in place of $\hat{J}$. The possible values of m_s are $\frac{1}{2}$ and $-\frac{1}{2}$. Let us denote

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the state with $s = \frac{1}{2}$ and $m_s = \frac{1}{2}$ (spin-up along z) by $|\frac{1}{2}, \frac{1}{2}\rangle$ and the state with $s = \frac{1}{2}$ and $m_s = -\frac{1}{2}$ (spin-down along z) by $|\frac{1}{2}, -\frac{1}{2}\rangle$, where we have started to put a comma between the quantum numbers j and m_j in $|j, m_j\rangle$ for clarity of notation, which we will often follow from now on. Because the spin has two possible components (up or down), the Hilbert space of a spin-half system is two-dimensional and complex. The basis vectors are

$$|\frac{1}{2}, \frac{1}{2}\rangle \equiv |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |\frac{1}{2}, -\frac{1}{2}\rangle \equiv |2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{4.6.6}$$

Let us construct the matrices for $\hat{S}_x$, $\hat{S}_y$, $\hat{S}_z$, and $\hat{\vec{S}}^2$. First, the matrices $\hat{S}_z$ and $\hat{\vec{S}}^2$ are diagonal, since they are the ones that are simultaneously diagonalized. The matrix elements are

$$\langle s' m_s' | \hat{\vec{S}}^2 | s m_s \rangle = s(s+1)\hbar^2 \delta_{s's} \delta_{m_s' m_s}, \qquad \langle s' m_s' | \hat{S}_z | s m_s \rangle = m_s \hbar \delta_{s's} \delta_{m_s' m_s} \tag{4.6.7}$$

In matrix form,

$$\hat{\vec{S}}^2 = \frac{3}{4}\hbar^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{4.6.8}$$

Using the matrix elements for $\hat{S}_\pm$, we have

$$\hat{S}_+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \hat{\vec{S}}^2 = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \tag{4.6.9}$$

Then, using $\hat{S}_x = \frac{1}{2}(\hat{S}_+ + \hat{S}_-)$ and $\hat{S}_x = \frac{1}{2i}(\hat{S}_+ - \hat{S}_-)$, we have

$$\hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \tag{4.6.10}$$

The spin angular momentum can be rewritten as

$$\vec{S} = \frac{\hbar}{2} \vec{\sigma}, \tag{4.6.11}$$

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where the components of $\vec{\sigma}$ are called the Pauli matrices, and given by

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{4.6.12}$$

Observe that

$$[\hat{S}_i, \hat{S}_j] = i\hbar\epsilon_{ijk}\hat{S}_k \tag{4.6.13}$$

and

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k. \tag{4.6.14}$$

A general state of a spin-half system is then given by

$$|\chi\rangle = a|\frac{1}{2}, \frac{1}{2}\rangle + b|\frac{1}{2}, \frac{1}{2}\rangle = \begin{pmatrix} a \\ b \end{pmatrix}, \tag{4.6.15}$$

where a and b are complex constants. Such 2-component state is called a spinor.

Again, spin is a purely quantum mechanical property without classical analogue. Particles such as the electron, the proton, and the neutron have a half spin. Quarks have a half spin too. Some particles, such as pions, have zero spin. The photon has spin one. Spin is one of the defining intrinsic property of particles. We distinguish two different particles such as the electron from the proton using the difference in their intrinsic properties of mass, charge, and spin. Every electron comes with the same mass, charge, and spin.

Exercise 4.6.2:

Show that the possible values of a measurement of the spin angular momentum along any direction could give only either $+\frac{\hbar}{2}$ or $-\frac{\hbar}{2}$ and nothing else.

Solution: Consider an arbitrary direction with unit vector $\hat{e}_a = a_x\hat{e}_x + a_y\hat{e}_y + a_z\hat{e}_z$. The spin vector operator along this direction is then $\hat{S}_a = \hat{\vec{S}} \cdot \hat{e}_a = a_x\hat{S}_x + a_y\hat{S}_y + a_z\hat{S}_z) = \frac{\hbar}{2}\begin{pmatrix} a_z & a_x - ia_y \\ a_x + ia_y & -a_z \end{pmatrix}$. The eigenvalues are obtained by solving $|\hat{S}_a - \lambda I| = \begin{vmatrix} \frac{\hbar}{2}a_z - \lambda & \frac{\hbar}{2}a_x - i\frac{\hbar}{2}a_y \\ \frac{\hbar}{2}a_x + i\frac{\hbar}{2}a_y & -\frac{\hbar}{2}a_z - \lambda \end{vmatrix} = 0$, which gives the quadratic equation $\lambda^2 - \frac{\hbar^2}{4}a_z^2 - \frac{\hbar^2}{4}a_x^2 - \frac{\hbar^2}{4}a_y^2 = 0$ and the solutions are $\lambda = \pm\frac{\hbar}{2}|\hat{e}_a| = \pm\frac{\hbar}{2}$, since $\hat{e}_a$ is a unit

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vector and $|\hat{e}_a| = 1$. Therefore, the eigenvalues of $\hat{S}_a$ along unit vector $\hat{e}_a$ is $\pm\frac{\hbar}{2}$. Let us write the corresponding state vectors as $|+\rangle$ and $|-\rangle$. To find the vector $|+\rangle$, we solve $\begin{pmatrix} a_z - 1 & a_x - ia_y \\ a_x + ia_y & -a_z - 1 \end{pmatrix}\begin{pmatrix} b \\ c \end{pmatrix} = 0$, which gives $(a_z - 1)b + (a_x - ia_y)c = 0$ and $(a_x + ia_y)b + -(a_z + 1)c = 0$. Using the latter, $\frac{c}{b} = \frac{a_x+ia_y}{a_z+1}$. To write the solutions in simpler form, let us writing the components of $\hat{a}$ using spherical coordinates: $a_x = \sin\theta\cos\phi$, $a_y = \sin\theta\sin\phi$, and $a_z = \cos\theta$. Then, $\frac{c}{b} = \frac{\sin\theta\cos\phi+i\sin\theta\sin\phi}{\cos\theta+1} = \frac{\sin\theta e^{i\phi}}{\cos\theta+1} = \frac{\sin\frac{\theta}{2}e^{i\phi}}{\cos\frac{\theta}{2}}$. Thus, we can choose, $b = \cos\frac{\theta}{2}$ and $c = \sin\frac{\theta}{2}e^{i\phi}$. Repeating similar steps for $|-\rangle$ too, we have

$$|+\rangle = \begin{pmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2}e^{i\phi} \end{pmatrix}, \qquad |-\rangle = \begin{pmatrix} -\sin\frac{\theta}{2}e^{-i\phi} \\ \cos\frac{\theta}{2} \end{pmatrix} \tag{4.6.16}$$

Exercise 4.6.3:

Determine the matrices for $\hat{\vec{L}}^2$, $\hat{L}_z$, and $\hat{L}_x$ for the four $n = 2$ states of hydrogen in the bases $|nlm_l\rangle = |200\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$, $|211\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$, $|210\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$, and $|21-1\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$

Exercise 4.6.4: Determine the matrices for $\hat{\vec{J}}^2$, $\hat{J}_z$, $\hat{J}_x$, and $\hat{J}_y$ for $j = \frac{3}{2}$ and their commutations.

Question 4.6.1: Can we have a scenario in which angular momentum is not quantized? Compare and contrast your answer with energy.

4.6.3 Magnetic moment of the electron

The relation between the magnetic moment and the spin angular momentum of the electron in quantum mechanics is given by

$$\vec{\mu} = g\frac{q}{2m}\vec{S}, \tag{4.6.17}$$

which has an additional factor of g called gyromagnetic factor (or simply g–value). The g–value is a dimensionless numerical factor. It's numerical value is discussed in Ex. 4.6.1. [A quantum mechanical calculation of the g–value requires relativistic quantum mechanics.]

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Question 4.6.2: How could the gyromagnetic factor be different from one, within classical physics itself?

For the electron, $q = -e$, and

$$\vec{\mu} = -g\frac{e}{2m}\vec{S}, \tag{4.6.18}$$

The corresponding operator,

$$\hat{\vec{\mu}} = -g\frac{e}{2m}\hat{\vec{S}} = -\frac{g}{2}\frac{e\hbar}{2m}\vec{\sigma} = -\frac{g}{2}\mu_B\vec{\sigma}, \tag{4.6.19}$$

where

$$\mu_B = \frac{e\hbar}{2m} \tag{4.6.20}$$

is called Bohr magneton.

4.6.4 Electron in a magnetic field

Consider an electron at rest in an external magnetic field $\vec{B}$. Because, the electron has an intrinsic magnetic moment due to its intrinsic spin angular momentum, the energy because of the interaction between its spin and the external magnetic field is given by

$$\hat{H} = \hat{V} = -\hat{\vec{\mu}} \cdot \vec{B}. \tag{4.6.21}$$

For a magnetic field along the $z-$axis, $\vec{B} = B\hat{z}$, and

$$\hat{H} = -\hat{\mu}_z B = -\left(-\frac{g}{2}\frac{e}{m}\hat{\vec{S}}\right) \cdot B\hat{z} = \frac{g}{2}\frac{eB}{m}\hat{S}_z = \omega_s \hat{S}_z = \frac{g}{2}\frac{eB\hbar}{2m}\sigma_z, \tag{4.6.22}$$

where

$$\omega_s = \frac{g}{2}\frac{eB}{m} = \frac{g}{2}\omega_c \tag{4.6.23}$$

is called spin precession (or Larmor) frequency and

$$\omega_c = \frac{eB}{m} \tag{4.6.24}$$

is called cyclotron frequency. The $g-$factor has an approximate value $g \approx 2$ (but not exactly). Therefore, the spin precession frequency ω_s is not equal to the cyclotron frequency. We rewrite the Hamiltonian as

$$\hat{H} = \omega_s S_z, \tag{4.6.25}$$

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In the bases in which $\hat{\vec{S}}^2$ and $\hat{S}_z$ are diagnonalized, the eigenstates are given by (4.6.6), and

$$\begin{aligned}\hat{H}|\frac{1}{2},\frac{1}{2}\rangle &= \omega_s\hat{S}_z|\frac{1}{2},\frac{1}{2}\rangle = \frac{1}{2}\hbar\omega_s|\frac{1}{2},\frac{1}{2}\rangle,\\ \hat{H}|\frac{1}{2},-\frac{1}{2}\rangle &= \omega_s\hat{S}_z|\frac{1}{2},-\frac{1}{2}\rangle = -\frac{1}{2}\hbar\omega_s|\frac{1}{2},-\frac{1}{2}\rangle.\end{aligned} \tag{4.6.26}$$

Therefore, the interaction of the spin of the electron with the external magnetic field leads to two energy levels. These energy levels correspond the spin-up state, $|\frac{1}{2},\frac{1}{2}\rangle$, and spin-down state, $|\frac{1}{2},-\frac{1}{2}\rangle$, with a gap of $\hbar\omega_s$ between them as shown in the figure below.

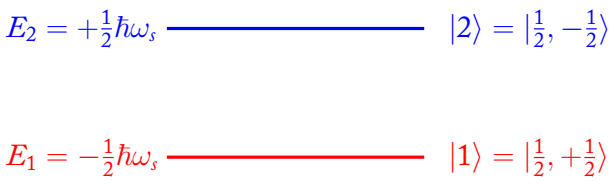


Figure 4.6.1: Energy levels spin-up and spin-down states of an electron in a magnetic field

Exercise 4.6.5: Show that the possible energy eigenvalues of energy for an electron in a magnetic field are $\pm\frac{1}{2}\hbar\omega_s$ regardless of the choice of direction of $\vec{B}$.

Solution: Consider a magnetic field along an arbitrary direction $\vec{B} = B_x\hat{e}_x + B_y\hat{e}_y + B_z\hat{e}_z$, the Hamiltonian operator for the interaction between the spin of the electron and the magnetic field is

$$\hat{H} = \left(\frac{g}{2}\frac{e}{m}\hat{\vec{S}}\right)\cdot\vec{B} = \frac{g}{2}\frac{e\hbar}{2m}\begin{pmatrix} B_z & B_x - iB_y \\ B_x + iB_y & -B_z \end{pmatrix} \tag{4.6.27}$$

The eigenvalues of the matrix are obtained from $\begin{vmatrix} B_z-\lambda & B_x - iB_y \\ B_x + iB_y & -B_z-\lambda \end{vmatrix} = 0$, which gives $\lambda = \pm B$, where $B = |\vec{B}|$. Therefore, the eigenvalues of $\hat{H}$ are $\pm\frac{g}{2}\frac{e\hbar B}{2m} = \pm\frac{1}{2}\hbar\omega_s$.

4.6.5 The Stern-Gerlach experiment

Question 4.6.3: What is the force acting on an electron in a constant magnetic field due to its spin?

The Stern-Gerlach experiment first performed in 1922, silver atoms are fired along an axis perpendicular to z and pass through a region of varying magnetic field $\vec{B} = B(z)\hat{z}$. The observed

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result is schematically shown in the figure below.

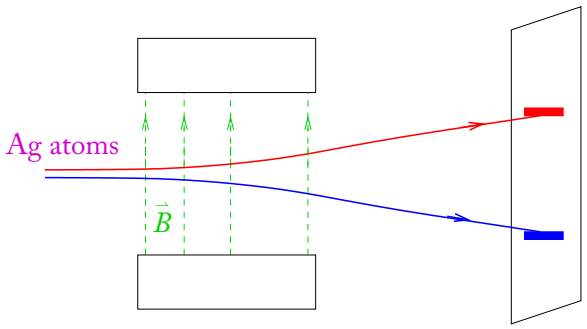


Figure 4.6.2: Stern-Gerlach deflection of silver atoms passing through non-unform magnetic field.

With a varying magnetic field, the force acting on an electron with spin-up is

$$\vec{F}_1 = -\vec{\nabla}V_1 = -\frac{1}{2}\hbar\vec{\nabla}\omega_s = -\frac{g}{2}\frac{e\hbar}{2m}\frac{\partial B(z)}{\partial z}\hat{z} \tag{4.6.28}$$

and the force acting on an electron with spin-down is

$$\vec{F}_2 = -\vec{\nabla}V_2 = \frac{1}{2}\hbar\vec{\nabla}\omega_s = \frac{g}{2}\frac{e\hbar}{2m}\frac{\partial B(z)}{\partial z}\hat{z} \tag{4.6.29}$$

Silver atom has one valence electron, and it is the spin of this electron that interacts with the external magnetic field. Thus, taking $\frac{\partial B(z)}{\partial z} < 0$, the force on a spin-up electron points up while that on a spin-down electron points down in the opposite direction. Therefore, the electrons are deflected up or down depending on the whether they are spin-up or spin-down along $\vec{B}$, consistent with the Stern-Gerlach experiment.

4.6.6 Spin precession

Now we want to take a look at the expectation value of the spin operator $\hat{\vec{S}}$ in an arbitrary state. Let us write the state at $t = 0$, $|\chi(0)\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$, where $|a|^2 + |b|^2 = 1$ and rewriting $a = \cos\frac{\alpha}{2}$ and $b = \sin\frac{\alpha}{2}$, as

$$|\chi(0)\rangle = \cos\frac{\alpha}{2}|\frac{1}{2}\frac{1}{2}\rangle + \sin\frac{\alpha}{2}|\frac{1}{2}-\frac{1}{2}\rangle = \begin{pmatrix} \cos\frac{\alpha}{2} \\ \sin\frac{\alpha}{2} \end{pmatrix}, \tag{4.6.30}$$

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Because each term is a stationary state,

$$|\chi(t)\rangle = e^{-\frac{i}{2}\omega_s t}\cos\frac{\alpha}{2}|\frac{1}{2}\frac{1}{2}\rangle + e^{\frac{i}{2}\omega_s t}\sin\frac{\alpha}{2}|\frac{1}{2}-\frac{1}{2}\rangle = \begin{pmatrix} e^{-\frac{i}{2}\omega_s t}\cos\frac{\alpha}{2} \\ e^{\frac{i}{2}\omega_s t}\sin\frac{\alpha}{2} \end{pmatrix}, \tag{4.6.31}$$

The expectation values for the $\hat{S}_z$ can be read off immediately, since each stationary state is an eigenstate of $\hat{S}_z$,

$$\langle\hat{S}_z\rangle = |e^{-\frac{i}{2}\omega_s t}\cos\frac{\alpha}{2}|^2\frac{\hbar}{2} - |e^{\frac{i}{2}\omega_s t}\sin\frac{\alpha}{2}|^2\frac{\hbar}{2} = \left(\cos^2\frac{\alpha}{2} - \sin^2\frac{\alpha}{2}\right)\frac{\hbar}{2} = \frac{\hbar}{2}\cos\alpha. \tag{4.6.32}$$

Alternatively, doing the math directly,

$$\begin{aligned}\langle\hat{S}_z\rangle &= \langle\chi(t)|\hat{S}_z|\chi(t)\rangle \\ &= \begin{pmatrix} e^{\frac{i}{2}\omega_s t}\cos\frac{\alpha}{2} & e^{-\frac{i}{2}\omega_s t}\sin\frac{\alpha}{2} \end{pmatrix}\frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} e^{-\frac{i}{2}\omega_s t}\cos\frac{\alpha}{2} \\ e^{\frac{i}{2}\omega_s t}\sin\frac{\alpha}{2} \end{pmatrix} = \frac{\hbar}{2}\cos\alpha\end{aligned} \tag{4.6.33}$$

Moreover,

$$\begin{aligned}\langle\hat{S}_x\rangle &= \langle\chi(t)|\hat{S}_x|\chi(t)\rangle \\ &= \begin{pmatrix} e^{\frac{i}{2}\omega_s t}\cos\frac{\alpha}{2} & e^{-\frac{i}{2}\omega_s t}\sin\frac{\alpha}{2} \end{pmatrix}\frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} e^{-\frac{i}{2}\omega_s t}\cos\frac{\alpha}{2} \\ e^{\frac{i}{2}\omega_s t}\sin\frac{\alpha}{2} \end{pmatrix} = \frac{\hbar}{2}\sin\alpha\cos\omega_s t\end{aligned} \tag{4.6.34}$$

and

$$\begin{aligned}\langle\hat{S}_y\rangle &= \langle\chi(t)|\hat{S}_y|\chi(t)\rangle \\ &= \begin{pmatrix} e^{\frac{i}{2}\omega_s t}\cos\frac{\alpha}{2} & e^{-\frac{i}{2}\omega_s t}\sin\frac{\alpha}{2} \end{pmatrix}\frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} e^{-\frac{i}{2}\omega_s t}\cos\frac{\alpha}{2} \\ e^{\frac{i}{2}\omega_s t}\sin\frac{\alpha}{2} \end{pmatrix} = \frac{\hbar}{2}\sin\alpha\sin\omega_s t.\end{aligned} \tag{4.6.35}$$

Thus, summarizing,

$$\langle\hat{S}_x\rangle = \frac{\hbar}{2}\sin\alpha\cos\omega_s t, \quad \langle\hat{S}_y\rangle = \frac{\hbar}{2}\sin\alpha\sin\omega_s t, \quad \langle\hat{S}_z\rangle = \frac{\hbar}{2}\cos\alpha, \tag{4.6.36}$$

the expectation values of the three components form the components of a vector in spherical coordinates with angle α from the $z-$axis and precessing around it with precession frequency

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ω_s as shown in the figure below for the case of a spin-up state.

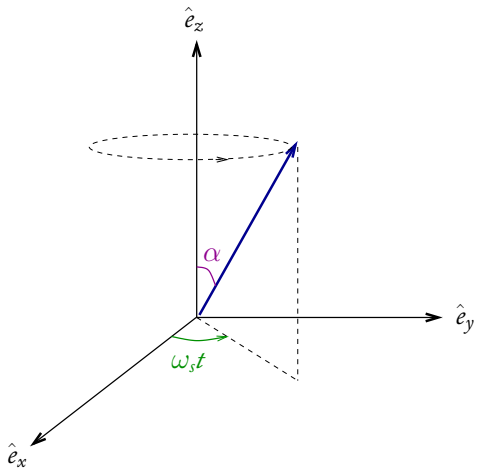


Figure 4.6.3: Spin precession.

This is called Larmor precession. The angle $\alpha \to \pi - \alpha$ for the case of a spin-down state. Thus, there are only two angles at which the spin precesses.

Note that the above analysis is for $\langle \hat{S}_x \rangle$, $\langle \hat{S}_y \rangle$, and $\langle \hat{S}_z \rangle$, which are just average values for each component, and the average values follow classical evolution. What about the eigenvalues themselves?

For a spin-up case, the eigenvalue of $\hat{S}_z$ is $\frac{\hbar}{2}$. We also know that the eigenvalue of $\hat{\vec{S}}^2$ is $\frac{\hbar}{2}(\frac{\hbar}{2}+1) = \frac{3\hbar^2}{4}$, or the magnitude of $|\hat{\vec{S}}^2|$ has a value of $\frac{\sqrt{3}\hbar}{2}$, which is greater than $\frac{\hbar}{2}$. Therefore, the vector cannot lie all along the $z-$axis. If it did, then that would be contradictory, since it would mean that the x and y components are exactly zero, and thus know in addition to the z component, which the corresponding operators do not commute and their values cannot be precisely known simultaneously. Moreover, the values of angular momentum along any fixed direction take only discrete $(2j+1)$ values, and this is often called space quantization. These properties hold two for higher spin or orbital angular momentum. As an example, the figure below shows the features for $j = 1$ with the component of $\hat{\vec{J}}$ along some arbitrary fixed direction $\hat{e}_a$ diagonalized.

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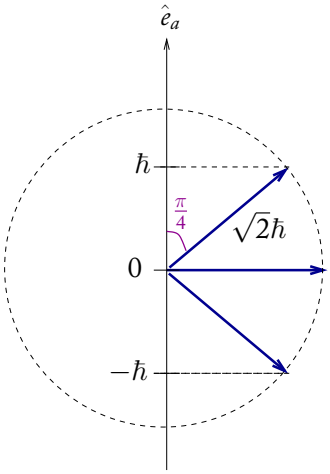


Figure 4.6.4: Space quantization for $j = 1$.

In the case when we choose the direction $\hat{e}_a = \hat{e}_z$, and considering $j = 1$ and $m_j = 1$, we have $|\vec{J}|^2 = 2\hbar^2$ and $|J_z| = \hbar$. We cannot tell the magnitudes for the other components $|J_x|$ and $|J_y|$, consistent with the noncommuting property of the components of angular momentum. But we know $|J_x|^2 + |J_y|^2 = |\vec{J}|^2 - |J_z|^2 = 2\hbar^2 - \hbar^2 = \hbar^2$.

4.7 Addition of angular momentum

Suppose we have two angular momenta represented by operators $\hat{\vec{J}}_1$ and $\hat{\vec{J}}_2$ with corresponding eigenstates $|j_1, m_{j_1}\rangle$ and $|j_2, m_{j_2}\rangle$ for the simultaneously diagonalizable set in each. The total angular momentum operator is $\hat{\vec{J}} = \hat{\vec{J}}_1 + \hat{\vec{J}}_2$.

> **Exercise 4.7.1:** Show that the components of $\hat{\vec{J}}$ obey the commutation relations of angular momentum given that the components of each of $\hat{\vec{J}}_1$ and $\hat{\vec{J}}_2$ do.
>
> **Solution:** $[\hat{J}_i, \hat{J}_j] = [\hat{J}_{1i} + \hat{J}_{2i}, \hat{J}_{1j} + \hat{J}_{2j}] = [\hat{J}_{1i}, \hat{J}_{1j}] + [\hat{J}_{1i}, \hat{J}_{1j}] = i\hbar\epsilon_{ijk}\hat{J}_{1k} + i\hbar\epsilon_{ijk}\hat{J}_{2k} = i\hbar\epsilon_{ijk}(\hat{J}_{1k} + \hat{J}_{2k}) = i\hbar\epsilon_{ijk}\hat{J}_k$.

Therefore, $\hat{\vec{J}}^2$ and $\hat{J}_z$ form a simultaneously diagonalizable set. Let the corresponding states be $|j, m_j\rangle$.

> **Question 4.7.1:** Which of the following are true about the relations among j, m_j, j_1, m_{j_1}, j_2, and m_{j_2}? (A). $j_{max} = j_1 + j_2$, (B). $m_{j_{max}} = j_1 + j_2$, (C). $j_{min} = -j_1 - j_2$, (D). $m_{j_{min}} = -j_1 - j_2$,

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4.7.1 Triplet and singlet states of a system of two spin-halves

Consider two electrons. Each electron can be in a spin-up or a spin-down state. What is the total spin angular momentum of the two-electron system? What do we get if we measure the spin of an individual electron? To get started, we know that each electron could be found either in state $|\frac{1}{2}, \frac{1}{2}\rangle$ or $|\frac{1}{2}, -\frac{1}{2}\rangle$. Thus, putting the state of the first electron as a first factor and the second as the second one in a direct product of states as

$$|j_1, m_{j_1}\rangle \otimes |j_2, m_{j_2}\rangle, \tag{4.7.1}$$

there are a total of four possible combinations for two electrons,

$$|\frac{1}{2}, \frac{1}{2}\rangle \otimes |\frac{1}{2}, \frac{1}{2}\rangle, \quad |\frac{1}{2}, \frac{1}{2}\rangle \otimes |\frac{1}{2}, -\frac{1}{2}\rangle, \quad |\frac{1}{2}, -\frac{1}{2}\rangle \otimes |\frac{1}{2}, \frac{1}{2}\rangle, \quad |\frac{1}{2}, -\frac{1}{2}\rangle \otimes |\frac{1}{2}, -\frac{1}{2}\rangle. \tag{4.7.2}$$

There are two states with unique values for the total spin, one with $m_j = \frac{1}{2} + \frac{1}{2} = 1$ and the other with $m_j = -\frac{1}{2} - \frac{1}{2} = -1$. Moreover, the total values of j ranges from largest value of m_j to smallest value of m_j in steps on unity. The state with $m_j = 1$ is

$$|1, 1\rangle = |\frac{1}{2}, \frac{1}{2}\rangle \otimes |\frac{1}{2}, \frac{1}{2}\rangle \tag{4.7.3}$$

There are two additional states with $j = 1$: $m_j = 0$ and $m_j = -1$. These are obtained by applying the lowering operator. In particular, the total angular momentum is

$$\hat{\vec{J}} = \hat{\vec{J}}_1 + \hat{\vec{J}}_2. \tag{4.7.4}$$

$\hat{\vec{J}}_1$ acts on the first factor and $\hat{\vec{J}}_2$ acts on the second factor in the direct produce of states. It is convenient to rewrite $\hat{\vec{J}}$ as

$$\hat{\vec{J}} = \hat{\vec{J}}_1 \otimes 1 + 1 \otimes \hat{\vec{J}}_2. \tag{4.7.5}$$

to emphasise that the first term acts on the first factor and leaves the second factor unaffected while the second term acts on the second factor and leaves the first factor unaffected in the direct product of states. For the raising and the lowering operators,

$$\hat{J}_\pm = \hat{J}_{1\pm} \otimes 1 + 1 \otimes \hat{J}_{2\pm}. \tag{4.7.6}$$

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Now, we apply $\hat{\vec{J}}_-$ on the left and $\hat{\vec{J}}_{1-} \otimes 1 + 1 \otimes \hat{\vec{J}}_{2-}$ on the right of the unique state $|1,1\rangle = |\frac{1}{2},\frac{1}{2}\rangle \otimes |\frac{1}{2},\frac{1}{2}\rangle$. We have

$$\hat{\vec{J}}_-|1,1\rangle = \hbar\sqrt{(1+1)(1-1+1)}\,|1,1-1\rangle = \hbar\sqrt{2}\,|1,0\rangle \tag{4.7.7}$$

and

$$\begin{aligned}\left(\hat{\vec{J}}_{1-} \otimes 1 + 1 \otimes \hat{\vec{J}}_{2-}\right)|\tfrac{1}{2},\tfrac{1}{2}\rangle \otimes |\tfrac{1}{2},\tfrac{1}{2}\rangle &= \hat{\vec{J}}_{1-}|\tfrac{1}{2},\tfrac{1}{2}\rangle \otimes |\tfrac{1}{2},\tfrac{1}{2}\rangle + |\tfrac{1}{2},\tfrac{1}{2}\rangle \otimes \hat{\vec{J}}_{2-}|\tfrac{1}{2},\tfrac{1}{2}\rangle \\ &= \hbar\sqrt{(\tfrac{1}{2}+\tfrac{1}{2})(\tfrac{1}{2}-\tfrac{1}{2}+1)}\,|\tfrac{1}{2},\tfrac{1}{2}-1\rangle \otimes |\tfrac{1}{2},\tfrac{1}{2}\rangle \\ &\quad + |\tfrac{1}{2},\tfrac{1}{2}\rangle \otimes \hbar\sqrt{(\tfrac{1}{2}+\tfrac{1}{2})(\tfrac{1}{2}-\tfrac{1}{2}+1)}|\tfrac{1}{2},\tfrac{1}{2}-1\rangle \\ &= \hbar\left(|\tfrac{1}{2},-\tfrac{1}{2}\rangle \otimes |\tfrac{1}{2},\tfrac{1}{2}\rangle + |\tfrac{1}{2},\tfrac{1}{2}\rangle \otimes |\tfrac{1}{2},-\tfrac{1}{2}\rangle\right). \end{aligned} \tag{4.7.8}$$

Equating (4.7.7) and (4.7.8),

$$\begin{aligned}|1,0\rangle &= \frac{1}{\sqrt{2}}\left(|\tfrac{1}{2},\tfrac{1}{2}\rangle \otimes |\tfrac{1}{2},-\tfrac{1}{2}\rangle + |\tfrac{1}{2},-\tfrac{1}{2}\rangle \otimes |\tfrac{1}{2},\tfrac{1}{2}\rangle\right) \\ &= \frac{1}{\sqrt{2}}\left(|\tfrac{1}{2},\tfrac{1}{2};\tfrac{1}{2},-\tfrac{1}{2}\rangle + |\tfrac{1}{2},-\tfrac{1}{2};\tfrac{1}{2},\tfrac{1}{2}\rangle\right), \end{aligned} \tag{4.7.9}$$

where we interchangeably use the notation of replacing the direct product by a semicolon in the same ket.

Next, we apply $\hat{\vec{J}}_-$ on the left and $\hat{\vec{J}}_{1-} \otimes 1 + 1 \otimes \hat{\vec{J}}_{2-}$ on the right of the state we have already constructed, $|1,0\rangle = \frac{1}{\sqrt{2}}\left(|\frac{1}{2},\frac{1}{2};\frac{1}{2},-\frac{1}{2}\rangle + |\frac{1}{2},-\frac{1}{2};\frac{1}{2},\frac{1}{2}\rangle\right)$. We have

$$\hat{\vec{J}}_-|1,0\rangle = \hbar\sqrt{(1+0)(1-0+1)}\,|1,0-1\rangle = \hbar\sqrt{2}\,|1,-1\rangle \tag{4.7.10}$$

and, noting that $\vec{J}_{1-}|\frac{1}{2},-\frac{1}{2}\rangle = \vec{J}_{2-}|\frac{1}{2},-\frac{1}{2}\rangle = 0$ and keeping only nonzero terms,

$$\begin{aligned}&\tfrac{1}{\sqrt{2}}\left(\hat{\vec{J}}_{1-}|\tfrac{1}{2},\tfrac{1}{2}\rangle \otimes |\tfrac{1}{2},-\tfrac{1}{2}\rangle + |\tfrac{1}{2},-\tfrac{1}{2}\rangle \otimes \hat{\vec{J}}_{2-}|\tfrac{1}{2},\tfrac{1}{2}\rangle\right) \\ &= \tfrac{1}{\sqrt{2}}\hbar\sqrt{(\tfrac{1}{2}+\tfrac{1}{2})(\tfrac{1}{2}-\tfrac{1}{2}+1)}\,|\tfrac{1}{2},\tfrac{1}{2}-1\rangle \otimes |\tfrac{1}{2},-\tfrac{1}{2}\rangle \\ &\quad + |\tfrac{1}{2},-\tfrac{1}{2}\rangle \otimes \hbar\sqrt{(\tfrac{1}{2}+\tfrac{1}{2})(\tfrac{1}{2}-\tfrac{1}{2}+1)}|\tfrac{1}{2},\tfrac{1}{2}-1\rangle \\ &= \tfrac{1}{\sqrt{2}}\hbar\left(|\tfrac{1}{2},-\tfrac{1}{2}\rangle \otimes |\tfrac{1}{2},-\tfrac{1}{2}\rangle + |\tfrac{1}{2},-\tfrac{1}{2}\rangle \otimes |\tfrac{1}{2},-\tfrac{1}{2}\rangle\right)\end{aligned}$$

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$$= \sqrt{2}\hbar|\tfrac{1}{2},-\tfrac{1}{2}\rangle \otimes |\tfrac{1}{2},-\tfrac{1}{2}\rangle. \tag{4.7.11}$$

Equating (4.7.10) and (4.7.12),

$$\begin{aligned} |1,-1\rangle &= |\frac{1}{2},-\frac{1}{2}\rangle \otimes |\frac{1}{2},-\frac{1}{2}\rangle \\ &= |\frac{1}{2},-\frac{1}{2};\frac{1}{2},-\frac{1}{2}\rangle, \end{aligned} \tag{4.7.12}$$

consistent with what we expected as there is only one way for getting a total angular momentum state of $|1,-1\rangle$ out of two spin-half states.

Thus, we have three states with the same $j = 1$ but different $m_j = 1, 0, 1$. We call these states spin-1 triplet.

Because we had a total of four possible states listed in (4.7.2) out of a product of two spin$-\frac{1}{2}$ states, and we have only three of them in the triplet, we need to construct a fourth one. This one must have total $m_j = 0$ and, thus, a linear combination of $|\frac{1}{2},\frac{1}{2}\rangle\otimes|\frac{1}{2},-\frac{1}{2}\rangle$ and $|\frac{1}{2},-\frac{1}{2}\rangle\otimes|\frac{1}{2},\frac{1}{2}\rangle$. It must also be orthogonal to the triplet states. In particular, we can read off the state that must be orthogonal to $|1,0\rangle$ is

$$\frac{1}{\sqrt{2}}\left(|\frac{1}{2},\frac{1}{2}\rangle \otimes |\frac{1}{2},-\frac{1}{2}\rangle - |\frac{1}{2},-\frac{1}{2}\rangle \otimes |\frac{1}{2},\frac{1}{2}\rangle\right), \tag{4.7.13}$$

Because the relationship between $\hat{J}_z$ and $\hat{J}_{1z} + \hat{J}_{2z}$ is linear, the eigenvalues of $\hat{J}_z$ follow the addition rule $m_j = m_{j_1} + m_{j_2}$. On the other hand, the eigenvalue j comes through $j(j+1)\hbar^2$ being the eigenvalue of $\hat{\vec{J}}^2$, and the relation between $\hat{\vec{J}}$ and $\hat{\vec{J}}_1 + \hat{\vec{J}}_2$ is not linear, we do not simply have $j = j_1 + j_2$. Let us check the value of j in the above state. Noting that $\hat{\vec{J}}^2 = (\hat{\vec{J}}_1 + \hat{\vec{J}}_2)^2 = \hat{\vec{J}}_1^2 + \hat{\vec{J}}_2^2 + 2\hat{\vec{J}}_1\cdot\hat{\vec{J}}_2 = \hat{\vec{J}}_1^2 + \hat{\vec{J}}_2^2 + 2\hat{J}_{1z}\hat{J}_{2z} + 2\hat{J}_{1x}\hat{J}_{2x} + 2\hat{J}_{1y}\hat{J}_{2y} = \hat{\vec{J}}_1^2 + \hat{\vec{J}}_2^2 + 2\hat{J}_{1z}\hat{J}_{2z} + \hat{J}_{1+}\hat{J}_{2-} + \hat{J}_{1-}\hat{J}_{2+}$. Rewriting,

$$\hat{\vec{J}}^2 = \hat{\vec{J}}_1^2 \otimes 1 + 1 \otimes \hat{\vec{J}}_2^2 + 2\hat{J}_{1z}\otimes\hat{J}_{2z} + \hat{J}_{1+}\otimes\hat{J}_{2-} + \hat{J}_{1-}\otimes\hat{J}_{2+} \tag{4.7.14}$$

Applying (4.7.14) on (4.7.13),

$$\begin{aligned} &\hat{\vec{J}}^2 \tfrac{1}{\sqrt{2}}\left(|\tfrac{1}{2},\tfrac{1}{2}\rangle \otimes |\tfrac{1}{2},-\tfrac{1}{2}\rangle - |\tfrac{1}{2},-\tfrac{1}{2}\rangle \otimes |\tfrac{1}{2},\tfrac{1}{2}\rangle\right) \\ &= \tfrac{1}{\sqrt{2}}\left(\hat{\vec{J}}_1^2|\tfrac{1}{2},\tfrac{1}{2}\rangle \otimes |\tfrac{1}{2},-\tfrac{1}{2}\rangle - \hat{\vec{J}}_1^2|\tfrac{1}{2},-\tfrac{1}{2}\rangle \otimes |\tfrac{1}{2},\tfrac{1}{2}\rangle\right) \\ &+\tfrac{1}{\sqrt{2}}\left(|\tfrac{1}{2},\tfrac{1}{2}\rangle \otimes \hat{\vec{J}}_2^2|\tfrac{1}{2},-\tfrac{1}{2}\rangle - |\tfrac{1}{2},-\tfrac{1}{2}\rangle \otimes \hat{\vec{J}}_2^2|\tfrac{1}{2},\tfrac{1}{2}\rangle\right) \\ &+\tfrac{2}{\sqrt{2}}\left(\hat{J}_{1z}|\tfrac{1}{2},\tfrac{1}{2}\rangle \otimes \hat{J}_{2z}|\tfrac{1}{2},-\tfrac{1}{2}\rangle - \hat{J}_{1z}|\tfrac{1}{2},-\tfrac{1}{2}\rangle \otimes \hat{J}_{2z}|\tfrac{1}{2},\tfrac{1}{2}\rangle\right) \end{aligned}$$

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$$
\begin{aligned}
&+\tfrac{1}{\sqrt{2}}\left(\hat{J}_{1-}|\tfrac{1}{2},\tfrac{1}{2}\rangle\otimes\hat{J}_{2+}|\tfrac{1}{2},-\tfrac{1}{2}\rangle-\hat{J}_{1+}|\tfrac{1}{2},-\tfrac{1}{2}\rangle\otimes\hat{J}_{2-}|\tfrac{1}{2},\tfrac{1}{2}\rangle\right)\\
&=\tfrac{1}{2}(\tfrac{1}{2}+1)\hbar^2\tfrac{1}{\sqrt{2}}\left(|\tfrac{1}{2},\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},-\tfrac{1}{2}\rangle-|\tfrac{1}{2},-\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},\tfrac{1}{2}\rangle\right)\\
&+\tfrac{1}{2}(\tfrac{1}{2}+1)\hbar^2\tfrac{1}{\sqrt{2}}\left(|\tfrac{1}{2},\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},-\tfrac{1}{2}\rangle-|\tfrac{1}{2},-\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},\tfrac{1}{2}\rangle\right)\\
&\quad-\tfrac{\hbar^2}{4}\tfrac{2}{\sqrt{2}}\left(|\tfrac{1}{2},\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},-\tfrac{1}{2}\rangle-|\tfrac{1}{2},-\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},\tfrac{1}{2}\rangle\right)\\
&\quad-\hbar\tfrac{1}{\sqrt{2}}\left(|\tfrac{1}{2},\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},-\tfrac{1}{2}\rangle-|\tfrac{1}{2},-\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},\tfrac{1}{2}\rangle\right)\\
&\quad=\left(\tfrac{1}{2}(\tfrac{1}{2}+1)\hbar^2+\tfrac{1}{2}(\tfrac{1}{2}+1)\hbar^2-2\tfrac{\hbar^2}{4}-\hbar^2\right)\\
&\quad\tfrac{1}{\sqrt{2}}\left(|\tfrac{1}{2},\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},-\tfrac{1}{2}\rangle-|\tfrac{1}{2},-\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},\tfrac{1}{2}\rangle\right)=0 \qquad .
\end{aligned}
\tag{4.7.15}
$$

Note that if we did the same calculation on the $|10\rangle$, reading off from the calculation for $|00\rangle$ above, and noting that only the sing of the last term changes,

$$
\begin{aligned}
&\hat{\vec{J}}^2\tfrac{1}{\sqrt{2}}\left(|\tfrac{1}{2},\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},-\tfrac{1}{2}\rangle-|\tfrac{1}{2},-\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},\tfrac{1}{2}\rangle\right)\\
&\quad=\left(\tfrac{1}{2}(\tfrac{1}{2}+1)\hbar^2+\tfrac{1}{2}(\tfrac{1}{2}+1)\hbar^2-2\tfrac{\hbar^2}{4}+\hbar^2\right)\\
&\quad\tfrac{1}{\sqrt{2}}\left(|\tfrac{1}{2},\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},-\tfrac{1}{2}\rangle-|\tfrac{1}{2},-\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},\tfrac{1}{2}\rangle\right)\\
=\quad&\left(2\hbar^2\right)\tfrac{1}{\sqrt{2}}\left(|\tfrac{1}{2},\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},-\tfrac{1}{2}\rangle-|\tfrac{1}{2},-\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},\tfrac{1}{2}\rangle\right).
\end{aligned}
\tag{4.7.16}
$$

Therefore, the state has total angular momentum quantum number of $j=0$. Together with $m_j=0$, we then have

$$
\begin{aligned}
|00\rangle &= \frac{1}{\sqrt{2}}\left(|\frac{1}{2},\frac{1}{2}\rangle\otimes|\frac{1}{2},-\frac{1}{2}\rangle-|\frac{1}{2},-\frac{1}{2}\rangle\otimes|\frac{1}{2},\frac{1}{2}\rangle\right)\\
&= \frac{1}{\sqrt{2}}\left(|\frac{1}{2},\frac{1}{2};\frac{1}{2},-\frac{1}{2}\rangle-|\frac{1}{2},-\frac{1}{2};\frac{1}{2},\frac{1}{2}\rangle\right),
\end{aligned}
\tag{4.7.17}
$$

There is only one state with $j=0$, and we call it a singlet state.

Summarizing, the triplet states ($j=1$ multiplet) are

$$
\begin{cases}
|1,1\rangle=|\tfrac{1}{2},\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},\tfrac{1}{2}\rangle\\
|1,0\rangle=\tfrac{1}{\sqrt{2}}\left(|\tfrac{1}{2},\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},-\tfrac{1}{2}\rangle+|\tfrac{1}{2},-\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},\tfrac{1}{2}\rangle\right)\\
|1,-1\rangle=|\tfrac{1}{2},-\tfrac{1}{2}\rangle\otimes|\tfrac{1}{2},-\tfrac{1}{2}\rangle
\end{cases}
\tag{4.7.18}
$$

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and the singlet state ($j = 0$ singlet) is

$$\left|00\right\rangle = \frac{1}{\sqrt{2}}\left(\left|\tfrac{1}{2},\tfrac{1}{2}\right\rangle \otimes \left|\tfrac{1}{2},-\tfrac{1}{2}\right\rangle - \left|\tfrac{1}{2},-\tfrac{1}{2}\right\rangle \otimes \left|\tfrac{1}{2},\tfrac{1}{2}\right\rangle\right) \tag{4.7.19}$$

We write the resulting total spin angular momenta $s = 1, 0$ out of s_1 and s_2 as

$$\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0. \tag{4.7.20}$$

We observe that the values of j range from $j = j_1 + j_2 = 1$ to $j = j_1 - j_2 = 0$.

For spin angular momentum, we interchangeably use the operator $\hat{\vec{S}}$ for $\hat{\vec{J}}$ as we mentioned earlier, and the quantum numbers s and m_s for j and m_j.

> **Exercise 4.7.2:** Check (or recheck) using $\hat{\vec{S}}^2$ and $\hat{S}_z$ operators that the triplet states have $s = 1$ and $m_s = 1, 0, -1$ and the singlet state has $s = m_s = 0$.

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